

Arduino-Based Fancy Lights Controller

Assemble yourself a portable, four-channel multi-mode light controller using an Arduino Uno board and very few external components



T.K. HAREENDRAN

The fancy lights controller described here is built around the Arduino (an Open Source single-board microcontroller) platform that can be purchased in pre-assembled hardware form. The circuit is nothing but a portable four-channel, multi-mode digital light controller, realised using very few external components. Four LEDs are made to glow in different sequences and patterns, controlled from the Arduino board (Fig. 1).

For this project, in addition to an Arduino UNO board, you need a USB cable (A to B), PC running the Arduino IDE (0022), components (four LEDs and 220-ohm resistors, a 10-kilo-ohm resistor and a switch), accessories (wires, connectors and battery), and the code or sketch for the MCU.

In the prototype, four 5mm red LEDs (LED1 through LED4) are used at the output. However, the same output lines can also be used to control high-voltage incandescent lamp strings with the help of suitable solidstate light switch modules. A push-to-on microswitch (S1) is the one and only output mode selector in the circuit!

The screenshot of fancy lights controller wiring generated using Open Source Fritzing software is shown in Fig. 2. The table shows the function of switch S1 and corresponding output on LEDs (LED1 through LED4).

Arduino

The Arduino board is an easy-to-use yet powerful single-board computer that has gained considerable traction in

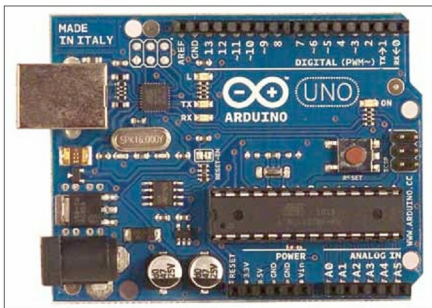


Fig. 1: Arduino board

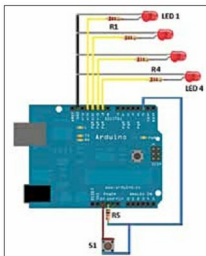


Fig. 2: Fritzing image of fancy lights controller

the hobby and professional market. It is Open Source, which means the hardware is reasonably priced and the development software free. An important feature of the Arduino is that you can create a control program on the host PC, upload it to the Arduino board and it will run automatically. This means that

LED1-LED4 Output w.r.t. Switch Action

Switch operation	LED output
Initial power 'on'	All 'off'
First push of S1	All 'on'
Second push of S1	Flashing
Third push of S1	Waving
Fourth push of S1	All 'off'

you connect the board to the host PC to develop and debug your program, but once that is done you no longer need the PC to run the program.

Uno is one of the series of USB Arduino boards. The Arduino Uno is a microcontroller board based on the ATmega328. It has 14 digital input/output pins (of which six can be used as PWM outputs), six analogue inputs, a 16MHz crystal oscillator, a USB connection, a power jack, an ICSP (in-circuit serial programming) header and a reset button. It contains everything needed to support the microcontroller; simply connect it to a computer through a USB